LARGE LANGUAGE MODELS IN HUMAN-MACHINE INTERACTION

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Abstract

This document serves as a template for the seminar write-up on Large Language Models in human-machine interaction. It will be replaced with the final abstract once the paper is complete.

Keywords: Large Language Models, Human-Machine Interaction, Seminar

CONTENTS

1 Introduction

In high-stakes environments such as aerospace, medicine, disaster response, military applications, and similar domains, the effectiveness and speed with which operators interact with the tools available to them are of critical importance. Traditionally, human–machine interaction (HMI) in such domains has been achieved through methods that have been field-tested over many years, such as user interfaces (UIs) utilizing buttons, joysticks, haptic-feedback sensors, and touchscreens [RSS+25]. In regulated application domains specific norms and regulations govern the permissible interaction mediums for HMI user interfaces, including standards such as IEC 60204-1 for industrial electrical equipment and NASA-STD-3001 for aeronautical systems.[Int16][CWF23]

However, with the introduction of transformer encoder–decoder models and the subsequent popularization of large language models such as Google's BERT, OpenAI's GPT-3, and more recently GPT-5, a wide range of new possibilities has emerged in the field [CYJ+25]. Moreover, recent advancements in large language models in the areas of speech-to-speech and visual language models have created additional and particularly interesting opportunities [CFD24].

This paper presents a literature review of recent advancements in relation to human–machine interaction within the context of safety-critical working environments.

2 Background

Before proceeding with the analysis of domain-specific work, a brief introduction to the involved technologies is necessary to provide the reader with sufficient background and contextual understanding. Accordingly, this section presents a concise overview of the relevant technologies.

In a human–machine interaction (HMI) system, the human can be regarded as a component equipped with four primary input–output channels: vision, hearing, touch, and motor activity (movement) [DFAB04]. Among these channels, vision and hearing are most commonly addressed in contemporary AI-driven interaction systems and are therefore particularly relevant in the context of large language model (LLM)–based interfaces. For the visual modality, current approaches primarily involve text-based language models and vision–language models that process textual input alongside visual information. For the auditory modality, interaction is typically realized through audio-based processing pipelines that combine speech recognition and speech synthesis with language models. The subsequent subsections further examine these modalities in more detail.

2.1 Text-Based LLM Interfaces

Since the introduction of the Transformer architecture in "Attention Is All You Need", text-based content generation and processing have become the central foundation of modern natural language processing (NLP).[VSP+17] In this context, large language models are a family of neural language models trained on vast amounts of textual data and capable of performing a wide range of natural language processing tasks, such as generating context-aware responses based on user-provided text prompts.[BMR+20] To utilize this family of models as a user interface, the model must be capable of understanding domain-specific context and of acting upon user prompts, rather than merely generating text responses. Several methods exist to enable this, a brief summary of these approaches is provided below.

Fine Tuning

Fine-tuning is the process of further training a pre-trained model on task- or domain-specific data by updating its existing parameters or, in some approaches, by introducing additional trainable parameters.[Mos23] By further training the model on a domain-specific dataset, for example technical data on extraterrestrial habitats, a base model such as GPT-5 can acquire contextual knowledge of the application domain and generate its responses accordingly.

However, a significant downside of this approach is the increased risk of model hallucinations. Hallucinations refer to the phenomenon in which a model generates incorrect or fabricated information in response to a given prompt.[GYA+24] Such behavior can pose a significant risk in safety-critical environments.

Retrieval Augmented Generation

Retrieval-Augmented Generation (RAG) introduces relevant external information to the language model without modifying the model's internal parameters.

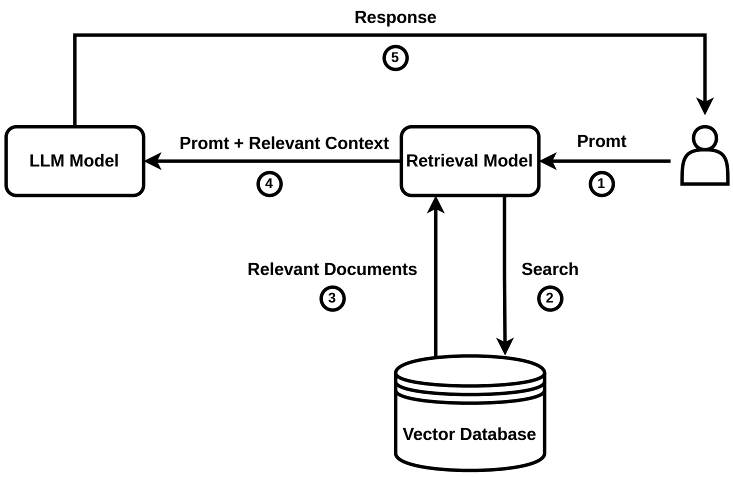


Figure 1: High-level RAG workflow that injects retrieved documents into the prompt instead of altering model weights.

As shown in Figure 1, the RAG workflow begins with a user prompt; however, instead of passing the prompt directly to the LLM, a similarity search is first performed over context-relevant information stored as embeddings in a vector database. The retrieved context is then provided together with the prompt to the LLM for grounded response generation.[BMR+20] Recent studies have shown this to be a more reliable method for context retrieval compared to fine-tuning.[OBME24]

However, the risk of hallucinations remains non-zero even with RAG; nevertheless, with well-designed context selection and retrieval strategies, this probability is significantly reduced, as demonstrated in numerous studies [JLF+24].

Tool Use / Function Calling

Tool calling, also referred to as function calling, is the ability of a large language model to invoke external functions by interpreting user context and generating structured Application Programming Interface (API) calls with the relevant parameters extracted from user prompts, as introduced by Schick et al. [SDYD+23].

In a practical setting, this means that the model generates structured tool calls, which are then used by a service layer to invoke HTTP requests to downstream services for task execution, as illustrated in Figure 2. By utilizing this method, the model can interact with external systems to trigger the execution of real-world functions.

In the context of human–machine interaction, this implies that a user can issue natural-language prompts to control robots and machines to execute a wide variety of tasks, as demonstrated by Vemprala et al. [VBBK23].

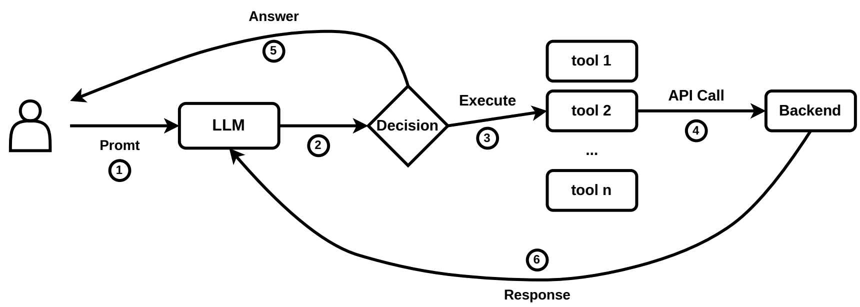


Figure 2: Conceptual workflow of tool (function) calling in a large language model.

Agentic Artificial Intelligence

What is commonly referred to as agentic AI is the combination of previously discussed methods to enable semi-autonomous decision-making systems. An AI agent typically integrates a large language model for reasoning and planning with tool-calling capabilities and memory in order to execute tasks without constant human supervision, requesting user input only when necessary to adjust its actions [YZY+23].

Such agents are currently most commonly used in programming-related tasks; a prominent example of an agentic AI system in this domain is OpenAI's Codex-based coding agent.

With regard to the use of such systems in human–machine interaction, Kim et al. note that the application of agentic systems in this context remains underexplored. However, their user study reveals that current agentic systems are still prone to failure, which caused usability issues for some participants [KLM24]. The agents in their experiment exhibited persistent hallucinations, leading some participants to doubt their own knowledge. Additionally, in cases where a physical robot embodiment was used to represent the agent, participants reported an uncanny-valley-like sense of unease during task execution. Nevertheless, the field and the underlying technology are still at an early stage of development, and user experience is expected to improve over time. The application of such systems in domains such as deep-space exploration remains a particularly promising research direction, as discussed by Heinicke et al. [FSG+21]. However, in their current state, the suitability of such systems for safety-critical applications remains uncertain.

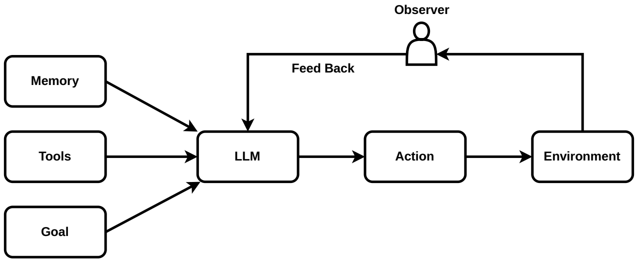


Figure 3: High-level agentic AI stack combining planning, memory, tool use, and execution loops.

2.2 Audio Based Interfaces

A critical component of human–machine interaction is speech, which represents one of the most natural and powerful means of human communication. In many HMI application domains, the ability to translate spoken user input into corresponding machine actions is of particular importance. Accordingly, this section discusses two methods that are currently employed to enable speech-driven interaction in such systems.

Speech-to-Text, LLM and Text-to-Speech

As highlighted by Haeb-Umbach et al. and other previous work, until recently this architecture represented the primary approach for providing voice support in interactive human–machine interaction (HMI) applications [HUWN+]. The overall pipeline typically consists of a speech detection and preprocessing layer, which forwards the audio signal to a speech-to-text (STT) model. The spoken input is thereby converted into text and subsequently processed by a large language model (LLM), which performs the reasoning and retrieval operations introduced in Section 2, such as retrieval-augmented generation (RAG). The LLM then produces a textual response, which is finally converted back into speech and presented to the user via a text-to-speech (TTS) system.

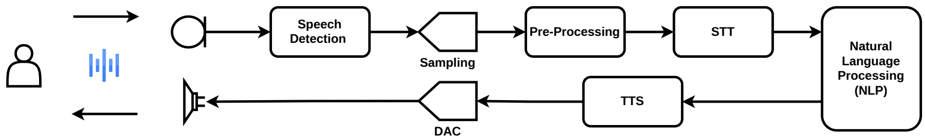


Figure 4: Signal chain for speech-to-text ingestion, LLM reasoning, and text-to-speech playback.

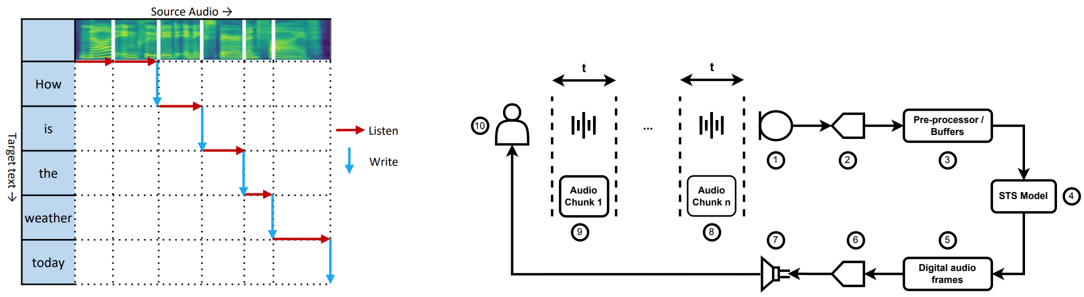
However, this pipeline also exhibits notable drawbacks. The sequential use of speech-to-text (STT), large language model (LLM) processing, and text-to-speech (TTS) synthesis can introduce perceptible latency. To mitigate this effect, some systems employ interaction design strategies such as playing short default utterances (e.g., “Let me think about that. . . ”) while the main response is being generated. Nevertheless, empirical user studies indicate that such waiting times can still negatively affect the naturalness and fluidity of human–machine interaction, particularly in conversational settings [FSG+21, KLM24].

On the positive side, such a configuration allows spoken input to be converted into textual form, which can facilitate integration with retrieval-augmented generation (RAG) systems and thereby support more efficient retrieval of contextually relevant information.

Speech-to-Speech LLMs

Direct end-to-end speech-to-speech models represent a relatively recent development within the broader family of large language and multimodal models. Unlike the cascading approach discussed in the previous section, these models do not rely on an explicit intermediate text representation; instead, linguistic processing is performed within a model trained directly on speech signals [SSJ+25]. This design can reduce end-to-end latency and mitigate error propagation that may arise from multiple processing stages in traditional multi-stage pipelines, and can support more natural conversational interaction with the system .

Within end-to-end speech-to-speech systems, two broad subcategories can be distinguished: direct models and real-time (streaming) models. Direct models process speech input as complete utterances, generating output only after the full input signal has been received. Their interaction pattern resembles cascading pipelines in terms of latency, as response generation is deferred until the entire input has been processed [ZFG+24].



(a) Streaming wait-k policy alternating between listening to additional source frames and emitting target tokens in real time [TLR+20].

(b) Streaming signal chain diagram with numbered legend.

Legend: 1) Microphone interface capturing the operator's raw speech
2) Low-latency Analog to Digital Converter
3) Pre-processing buffers that normalize, filter, and batch frames
4) Streaming STS model performing NLP task
5) Generated digital audio frames for the target utterance
6) Vocoder/decoder producing analog waveforms
7) Speaker/headset driver for immediate playback
8 - 9) The Audio is processed in chunks that are (t) in length, usually in the order of 20-40ms
10) End user simultaneously providing input and receiving translated speech.

Figure 5: Real-time speech-to-speech pipeline: (a) wait-k listen/write policy and (b) corresponding signal path.

In contrast, more recent real-time speech-to-speech systems, such as OpenAI's real-time speech model family adopt a streaming-based approach. Incoming speech is first digitized (e.g., using pulse-code modulation formats such as PCM16) and segmented into short audio frames on the order of tens of milliseconds. These segments are streamed incrementally to the model, which buffers a limited temporal context before beginning to generate spoken output [Ope24].

In Figure 5a, an example of such a system is illustrated using the simultaneous speech model introduced by Tan et al., showing how these models interleave listen and write operations according to a wait-k policy, allowing target-language output to begin before the full source utterance is observed [TLR$^+$20]. To ground this abstract description in the physical signal chain used in safety-critical HMI deployments, Figure 5b annotates a representative streaming speech-to-speech path from the operator's microphone through synthesis and playback.

This streaming formulation enables incremental processing and response generation, which can reduce perceived latency and support a more natural conversational flow compared to full utterance-level processing approaches. However, a downside of such an approach is that end-to-end speech-to-speech models typically require larger computational resources compared with cascaded systems discussed previously, and they can face challenges in generalizing across diverse domains due to model architecture. Additionally, as of late 2025, native retrieval-augmented generation (RAG) compatibility remains an active area of research for end-to-end speech models, with most practical RAG implementations still relying on intermediate text representations for effective document retrieval [MML$^+$25].

2.3 Vision-Language Models

Vision is another input channel that is of central importance to human psychological perception and interpretation, particularly in tasks where spatial reasoning and situational awareness are critical. Vision–language models extend machine interaction capabilities by enabling the joint processing of visual inputs and natural language, thereby allowing contextual understanding and reasoning grounded in visual information. These models exist at the intersection between computer vision and natural language processing [LWD$^+$25]. For example, such models can be applied to the analysis of human facial expressions to infer affective states from visual data. An illustrative example of this type of application is provided in Appendix **??**.

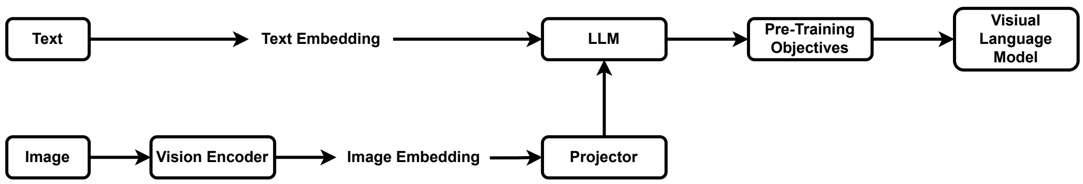


Figure 6: Conceptual VLM workflow: the vision encoder produces embeddings that are aligned and fused with LLM tokens to support multimodal reasoning and grounded responses.[LWD$^+$25]

Vision-Language Models (VLMs) work by using a vision encoder to convert images into numerical feature representations and a large language model (LLM) to convert text into contextual embeddings. The vision encoder extracts visual features, which are then transformed into a compatible format before being integrated with the language model's input. In many modern VLMs, a pre-trained LLM serves as the backbone, and a projection or fusion layer is used to inject visual embeddings into the LLM so it can process visual information alongside text. This enables multimodal understanding and generation. A representation of this process is provided in Figure 6 [LWD+25]. In the context of human–machine interaction (HMI), such models can be deployed alongside audio- and text-based models to enhance the overall user experience in interactive systems.

3 Domain Case Studies

In the previous section, the architectural foundations underlying the use of large language models (LLMs) in the context of human–machine interaction were introduced, encompassing text-based, audio-based, and vision–language modalities. This section builds on that foundation by reviewing selected studies from several safety-critical application domains; namely aerospace and remote sciences, emergency and environmental response, and industrial operations, with a focus on how LLM-based technologies are applied in human–machine interaction within these contexts.

3.1 Aerospace and Remote Science

Aerospace, Deep Sea wayages etc.

Psychological and Cognitive Impacts

3.2 Emergency and Environmental Response

3.3 Industrial Operations

4 Comparative Analysis and Metrics

5 Safety, Trust, and Deployment Constraints

6 Conclusion

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